

OCA BEU — Test Plan

SMC Bus Error Unit — Test Plan

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SMC Bus Error Unit — Test Plan

Verification strategy for the SystemC/TLM-2.0 LT model in `smc/peripherals/beu`. Benches: `test/beu_tb.cpp` (primary) and `test/beu_neg_tb.cpp` (negative / edge). Both self-check and print `ALL TESTS PASSED`.

1. Objectives

- Confirm the register map (offsets, masks, reset values) matches `bus_error_unit.rdl`.
- Confirm error recording semantics: accrued status, first-error latching into `CAUSE / PHYS_ADDR`, and recording-enable gating.
- Confirm dual interrupt delivery: local vs. PLIC lines, per-source masking, and software acknowledge / re-arm.
- Confirm robust bus behaviour: TLM response codes for malformed accesses.
- Achieve > 95 % line coverage of `src/beu.cpp` (achieved: 100 %).

2. Environment

- Accellera SystemC 2.3.x / 3.0.x + TLM-2.0, SystemC CCI 1.0, C++20.
- A tiny TLM initiator (`driver`) with a `tlm_quantumkeeper` issues 64-bit reads/writes and `transport_dbg` peeks; the DUT's interrupt outputs are wired to `sc_signal<bool>` observed by the bench.
- Error sources are driven through the `inject_error()` back door.

3. Test-bench architecture

Bench	Focus
<code>beu_tb</code>	Functional happy-path: reset, register RW/RO/masking, error injection, latching/priority, gating, interrupts, CCI.
<code>beu_neg_tb</code>	Error / edge branches: TLM response codes, decode miss, misalignment, <code>transport_dbg</code> edges, RO write-ignore, gated injection.

4. Primary test list (beu_tb)

#	Test	Checks
1	Reset values	CAUSE=0, PHYS_ADDR=0, ENABLE=0xE6, PLIC_ENABLE=0, ACCRUED=0, LOCAL_ENABLE=0, both IRQs low.
2	Register RW / RO / masking	ENABLE / PLIC_ENABLE / LOCAL_ENABLE mask reserved bits to 0xE6; CAUSE keeps [2:0]; PHYS_ADDR write ignored.
3	Basic injection	DCACHE_UNCORRECTABLE sets ACCRUED[7], CAUSE=7, PHYS_ADDR=addr.
4	First-error latch / re-arm	Second (ICACHE_TLBUS) error accrues but does not overwrite CAUSE / PHYS_ADDR; after CAUSE=0 + ACCRUED=0, next error latches CAUSE=1.
5	Recording-enable gating	With ENABLE=0, an error still sets accrued status but CAUSE / PHYS_ADDR stay clear.
6	Local + PLIC aggregation	LOCAL_ENABLE mask raises irq_local only; adding PLIC_ENABLE also raises irq_plic; clearing ACCRUED drops both.
7	Per-source mask selectivity	A masked-out source does not raise irq_plic; an unmasked source does.
8	transport_dbg / dbg_reg peek	Peeks return values without disturbing state; unmapped offset peek returns 0.
9	CCI introspection	access_delay_ns preset visible + mutable via broker.

5. Negative test list (beu_neg_tb)

#	Test	Expected
1	Ignore command	TLM_COMMAND_ERROR_RESPONSE
2	Wrong width (4 bytes)	TLM_BURST_ERROR_RESPONSE
3	Out-of-window (0x1000)	TLM_ADDRESS_ERROR_RESPONSE
4	Misaligned (0x4)	TLM_ADDRESS_ERROR_RESPONSE
5	In-window decode miss (0x30, read + write)	TLM_ADDRESS_ERROR_RESPONSE
6	Valid ENABLE read	TLM_OK_RESPONSE
7	RO PHYS_ADDR write	accepted (OK) but value unchanged (0)
8	transport_dbg bad length / misaligned / out-of-window / ignore cmd / unmapped write	returns 0
9	transport_dbg valid read/write of ENABLE	returns 8; value updates
10	Gated injection	with ENABLE=0, CAUSE stays 0 while ACCRUED sets.

6. Coverage

Run:

```
./run_tests.sh --coverage
```

`src/beu.cpp` reaches 100 % line coverage across the two benches. The primary bench covers the functional lines and the `dbg_reg` peek fall-through; the negative bench covers the four TLM error branches and the in-window decode-miss path.

7. Pass / fail criteria

- Each bench increments a failure counter on any failed `EXPECT_*` and prints `ALL TESTS PASSED` iff the counter is zero (exit code 0).
- CTest matches `ALL TESTS PASSED` / `FAILURE(S)` via the regexes in `test/CMakeLists.txt`.
- CI regression is driven by `smc/run_all_smc_tests.sh beu` (Release, ASAN, Coverage, CTest stages).